

Lab 3

Component Modelling & Architectural Pattern Selection

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SCENARIO: Problem Statement #62 – Food Bank Surplus Redistribution Platform (Sustainability & Green Tech)

1. Architectural Choice:

We chose a Microservices-Based Component Architecture for the Food Bank Surplus Redistribution Platform.

2. Scenario-Related Reasons:

Reason 1 (Spike Resilience & Selective Scalability):

Food donation postings occur sporadically with high batch volumes at commercial closing hours. The NotificationService and MatchingService must scale dynamically to broadcast real-time alerts within a 5 km radius without starving or degrading core listing and reservation endpoints.

Reason 2 (Decoupled Time-Critical Business Logic):

The platform mandates perishable expiry countdown timers. Decoupling the ExpiryTrackerService allows autonomous monitoring and batch claim revocation on a scheduled loop without introducing processing latency or blocking interactive database transactions conducted by shelters and donors.

3. Security Advantage:

Isolating regulatory donor licenses and verified shelter non-profit credentials inside a dedicated VerificationService component mitigates systemic risk. External actors interact strictly via scoped, ephemeral tokens (IAuthVerify), ensuring that an exploit in public batch reservation endpoints does not expose internal compliance records, cryptographic keys, or verification PINs.

4. Performance Benefit:

Asynchronous event-driven communication (INotificationTrigger) offloads compute-heavy geospatial matching (5 km radius spatial bounding box) from synchronous HTTP API transactions. When a donor registers food, the batch is committed in sub-second time, while push notifications propagate concurrently across background workers without degrading API response latency.

Component Diagram

